

基于目标检测的智能机器人动态视觉 SLAM 系统设计

摘要

随着人工智能技术在移动机器人与自动驾驶领域的深入应用，传统视觉 SLAM 系统在复杂动态场景下极易受运动物体干扰而导致定位失效。针对此痛点，本文提出了一种融合深度学习视觉感知与几何运动校验的智能化 SLAM 应用方案。该方案首先引入 YOLO 轻量级目标检测网络作为 AI 感知前端，实时提取环境的高层语义信息并锁定潜在的动态干扰源；其次，在 AI 语义先验的引导下，结合稀疏光流追踪与对极几何卡方检验，对目标边界框内的特征点进行智能化的运动校验与深度清洗，有效解决了目标框内静态背景被误剔除的问题；最后，利用结构约束对位姿进行补偿计算。实验表明，这种“AI 语义感知 + 底层几何校验”的系统架构，不仅显著提升了动态干扰环境下的全局定位精度与鲁棒性，同时保证了算法的实时性，为智能服务机器人在复杂现实场景中的落地应用提供了可靠的底层导航支持。

关键词：人工智能；视觉 SLAM；深度学习；YOLO；动态场景；系统集成

Abstract

With the deep application of artificial intelligence in mobile robots and autonomous driving, traditional visual SLAM systems are highly susceptible to moving objects in complex dynamic environments, leading to localization failures. To address this issue, this paper proposes an intelligent SLAM scheme combining deep learning visual perception with geometric motion validation. The scheme first introduces the lightweight YOLO object detection network as the AI perception frontend to extract high-level semantic information of the environment and lock potential dynamic interferences in real-time. Secondly, guided by the AI semantic prior, sparse optical flow tracking and epipolar geometry-based Chi-square test are integrated to perform intelligent motion validation and deep cleaning of feature points within target bounding boxes, effectively solving the problem of false elimination of static background points. Finally, structural constraints are utilized to compensate for the pose estimation. Experimental results show that the "AI semantic perception + low-level geometric validation" architecture significantly improves global localization accuracy and robustness in dynamic environments while maintaining real-time

performance, providing reliable underlying navigation support for intelligent service robots in complex real-world scenarios.

Keywords: Artificial Intelligence; Visual SLAM; Deep Learning; YOLO; Dynamic Environments; System Integration

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